MIL-F-45687(Ord)
16 February 1961
SUPERSEDING
FCP 981-58A
2 April 1958

MILITARY SPECIFICATION

FIRE CONTROL ELECTRONIC CALIBRATION STANDARDS: PACKAGING OF

1. SCOPE

1.1 Scope.- This specification covers the packaging, packing, and marking of electronic test equipment, calibration standards and associated equipment for all levels of protection (see 6.1).

1.2 Classification.- The equipment shall be classified in one of the following groups:

Group I	Items 60 lbs. to 150 lbs.	(See Figures 1 and 2)
Group II	Items 20 lbs. to 60 lbs.	(See Figures 1 and 2)
Group III	Items up to 20 lbs.	(See Figures 3, 4, 5 and 6)
Group IV	Items up to 20 lbs.	(See Figure 7)

1.2.1 The equipment shall be prepared for storage or shipment by one of the following levels of packaging and packing.

(a) Packaging

(1) Level A - Packaging which will afford adequate protection against corrosion deterioration or physical damage during shipment, handling, indeterminate storage, and world-wide distribution.

(2) Level C - Packaging which will afford adequate protection against physical damage during shipment to and from calibration depots for immediate use.

(b) Packing

(1) Level A and B - Packing which will afford adequate protection during shipment, handling, indeterminate storage, and permit domestic and world-wide distribution.

(2) Level C - Packing which will afford protection during shipment to and from calibration depots for immediate use.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

FSC 1290

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SPECIFICATIONS

Federal

C-F-206	Felt, Sheet, Wool, Pressed
FF-S-111	Screws; Wood
NN-P-515	Plywood, Container Grade
QQ-S-781	Strapping, Flat; Steel
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber
PPP-B-640	Boxes, Folding, Fiberboard, Corrugated, Triple Wall
PPP-C-843	Cushioning Material, Cellulosic
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced

Military

MIL-A-101	Adhesive, Water-Resistant, for Sealing Fiberboard Boxes
MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Flexible (Waterproofed)
MIL-B-131	Barrier Material; Water Vaporproof, Flexible
MIL-C-7769	Cushioning Material, Bound
MIL-I-45208	Inspection Requirements; General Specification For

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- Material shall be as specified herein. Material not definitely specified shall be of a good commercial quality, entirely suitable for the purpose. Material shall be free from defects and imperfections that might affect the serviceability of the finished preservation, packaging, and packing.

3.2 Preparation for packaging

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3.2.1 Interior of equipment.- The equipment shall be opened by instrument personnel to make certain that all tubes are positioned firmly in their bases, and shields and clips are firmly secured. Wherever possible, any unsupported assemblies inside the item shall be blocked, braced or supported by means of corrugated pads or die-cut inserts.

3.2.2 Exterior of equipment.- Cord assemblies which are attached to the item shall be coiled in such a manner as to prevent kinking and secured in position with twine. Cables, probes, dividers, adapters, etc. shall be individually wrapped with barrier material conforming to grade A, class 1 of Specification MIL-B-121 and overwrapped with paperboard conforming to type II of Specification PPP-P-291. The overwrapped component shall be secured to the instrument. All fasteners, snapslides, locking screws, studs, and moving parts shall be secured and tightened, and locked in position. Waveguide openings shall be covered with plastic dust caps or wrapped with barrier material conforming to class 1, type 1, grade A of Specification MIL-B-121. Items that are to be mounted on a base shall have the leveling screws removed, placed in a cotton cloth bag, and secured to the unit.

3.3 Preservation and packaging.

3.3.1 Level A.

3.3.1.1 Cleaning.- The item shall be cleaned by process C-1 of Specification MIL-P-116.

3.3.1.2 Unit packaging.

3.3.1.2.1 Group I.- The equipment shall be packaged method IA-14 of Specification MIL-P-116 as specified herein and as illustrated on figure I (for cabinet mounted) and figure 2 (for rack mounted).

3.3.1.2.1.1 Die cuts, pads, cells, sleeves, and panels shall provide sufficient clearance for knobs, dials, handles, and protruding parts of the unit. Fiberboard material shall be triple wall as specified in Specification PPP-B-640 or solid fiberboard type II, class 1 of Specification PPP-B-636 (550 P.S.I.). Plywood shall conform to type II or III of Specification NN-P-515. Felt pads shall conform to classification number 8R5 of Specification C-F-206. The inner container shall be reinforced with strips of pressure sensitive tape conforming to type III of Specification PPP-T-97. Closure shall be with pressure sensitive tape conforming to Specification PPP-T-76. The corners of the container shall be blunted prior to placing in a vaporproof bag fabricated from barrier material conforming to class 1 of Specification MIL-B-131. Closure of the barrier after the excess entrapped air is removed shall be by heat sealing.

3.3.1.2.1.2 The barrier enclosed inner container shall be cushioned on the top, bottom, sides and ends in a wood container conforming to style 4, class 2 of Specification PPP-B-621 or style A or B, overseas type of Specification PPP-B-601. Cushioning (a minimum of 6 inches thick) shall conform to type IV of Specification MIL-C-7769. For items weighing over 100 pounds, additional cushioning shall be proportionately provided to render the unit immobile. Closure of wood container shall be with slotted wood screws conforming to Specification FF-S-111 to permit reuse of the container.

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3.3.1.2.2 Group II.- The equipment shall be packaged method IA-14 of Specification MIL-P-116, as specified herein and as illustrated on figure I (for cabinet mounted) and figure 2 (for rack mounted).

3.3.1.2.2.1 Die cuts, pads, cells, sleeves and panels shall provide sufficient clearance for knobs, dials, handles, and protruding parts of the unit. Fiberboard material shall be double wall, type I, class 1 of Specification PPP-B-636. Plywood shall conform to type II or III of Specification NN-P-515. Felt pads shall conform to classification number 8R5 of Specification C-F-206. The inner container shall be reinforced with strips of pressure sensitive tape conforming to type III of Specification PPP-T-97. Closure shall be with pressure sensitive tape conforming to Specification PPP-T-76. A fiberboard box conforming to type 1, class 1, style RSC of Specification PPP-B-636 may be used in lieu of the sleeve shown in figure 1 when the interior sides of the box are such as to protect the item in the same manner as the sleeve. The corners of the container shall be blunted prior to placing in a vaporproof bag fabricated from barrier material conforming to class 1 of Specification MIL-B-131. Closure of the barrier after the excess entrapped air is removed shall be by heat sealing.

3.3.1.2.2.2 The barrier enclosed inner container shall be cushioned on the top, bottom, sides and ends in a wood container conforming to style 4, class 2 of Specification PPP-B-621 or style A or B, overseas type of Specification PPP-B-601. Cushioning (a minimum of 4 inches thick) shall conform to type IV of Specification MIL-C-7769. Closure of wood container shall be with slotted wood screws conforming to Specification FF-S-111 to permit reuse of the container.

3.3.1.2.3 Group III.- The equipment shall be packaged method IA-14 of Specification MIL-P-116, as specified herein and as illustrated on figures 3, 4, 5 and 6 as applicable. Die cuts, pads, cells, liners, sleeves and panels shall provide sufficient clearance for knobs, dials, handles and protruding parts of the unit.

3.3.1.2.3.1 Inner containers

3.3.1.2.3.1.1 Figure 3. The inner container shall be fabricated from double faced fiberboard material conforming to type I, class 1 of Specification PPP-B-636. The inner container shall be reinforced with strips of pressure sensitive tape conforming to Specification PPP-T-97. Filler pads and cells shall be secured to fiberboard panels with pressure sensitive tape conforming to Specification PPP-T-76 or adhesive conforming to Specification MIL-A-101. A fiberboard box may be used in lieu of a sleeve when the interior sides of the box are such as to protect the item in the same manner as the sleeve.

3.3.1.2.3.1.2 Figure 4. The inner container shall be of a fiberboard box conforming to type I, class I of Specification PPP-B-636. The unit shall be immobilized in the container by means of a hold down liner fabricated from double wall fiberboard conforming to type I, class 1 of Specification PPP-B-636.

3.3.1.2.3.1.3 Figure 5. The unit mounted on a cleated plywood base shall be immobilized in a fiberboard container by means of a hold down liner fabricated from double wall fiberboard conforming to type I, class 1 of Specification PPP-B-636. The fiberboard box shall conform to type I, class 1 of Specification PPP-B-636. A minimum clearance of 1 inch shall be provided between the unit and the sides, ends and top of container.

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3.3.1.2.3.1.4 Figure 6. The encased unit shall be wrapped with paperboard conforming to type II of Specification PPP-B-291.

3.3.1.2.3.1.5 Closure.- Closure of inner container or wrapped unit shall be with pressure sensitive tape conforming to Specification PPP-T-76.

3.3.1.2.3.2 Barrier.- The corners of the inner container shall be blunted prior to placing in a vaporproof bag fabricated from material conforming to class 1 of Specification MIL-B-131. Closure of the barrier after the excess entrapped air is removed shall be by heat sealing.

3.3.1.2.3.3 Outer container.- The barrier enclosed container shall be cushioned on the top, bottom, ends and sides in a fiberboard container conforming to type I, class 2, V3c of Specification PPP-B-636. Cushioning (a minimum of 2 inches thick) shall conform to type IV of Specification MIL-C-7769. Closure of the outer container shall be with pressure sensitive tape conforming to Specification PPP-T-76.

3.3.1.2.4 Group IV.- The equipment shall be packaged method IA-8 of Specification MIL-P-116, as specified herein and as illustrated on figure 7.

3.3.1.2.4.1 Die cuts, pads, cells or collars as applicable shall provide sufficient clearance for knobs, dials, switches and protruding parts of the unit. Item shall be wrapped with barrier material conforming to class I, grade A of Specification MIL-B-121 and overwrapped with cellulosic cushioning material (minimum 1/2 thickness) conforming to type III, class B of Specification PPP-C-843. The overwrapped item shall be placed in a bag fabricated from material conforming to class I of Specification MIL-B-131 and heat sealed. The bagged item shall be immobilized in a fiberboard container conforming to type I, class 2, W5c, style RSC or OPF of Specification PPP-B-636. Closure of the fiberboard container shall be with pressure sensitive tape conforming to Specification PPP-T-76.

3.3.2 Level C.- Packaging for groups I, II, III and IV shall be as specified in level A except the vaporproof barrier material shall be omitted. For groups I and II the outer container shall conform to style 4, class 1 of Specification PPP-B-621; style A or B, domestic type of Specification PPP-B-601.

3.4 Packing

3.4.1 Level A and B

3.4.1.1 Groups I and II.- The exterior container specified in 3.3.1.2.1.2 and 3.3.1.2.2.2 shall constitute the unit package and shipping container. The wood container shall be strapped in accordance with the appendix of the applicable box specification. Stapling of straps shall be omitted.

3.4.1.2 Groups III and IV.- A quantity of unit containers shall be immobilized in a shipping container conforming to style 4, class 2 of Specification PPP-B-621; style A or B, overseas type of Specification PPP-B-601 or type 1, class 2, V3c of Specification PPP-B-636. The gross weight of the wood shipping container shall not exceed 200 pounds. The wood shipping container shall be strapped in accordance with the appendix of the applicable box specification. The gross weight and size of the

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fiberboard shipping container shall not exceed the limitations of the box specification. In lieu of steel strapping, the fiberboard box shall be reinforced with tape conforming to type IV of Specification PPP-T-97. Closure shall be with tape conforming to Specification PPP-T-76 applied to all seams.

3.4.1.2.1 Alternate methods for groups III and IV.- The outer container of the unit package may be used as a shipping container provided the fiberboard box is reinforced with pressure sensitive tape conforming to type IV of Specification PPP-T-97. The pressure sensitive tape specified herein is used in lieu of steel strapping.

3.4.2 Level C.

3.4.2.1 Groups I and II.- The exterior container specified in paragraph 3.3.2, shall constitute the unit package and shipping container. The shipping container shall be strapped with steel strapping conforming to type I, class A or B of Specification QQ-S-781. Stapling of straps shall be omitted.

3.4.2.2 Groups III and IV.- Packing shall be as specified in paragraphs 3.4.1.2, except the shipping container shall conform to style 4, class 1 of Specification PPP-B-621; style A or B, domestic type of Specification PPP-B-601 or type I, class I of Specification PPP-B-636.

3.4.2.2.1 Alternate method for groups III and IV shall be the same as specified in paragraphs 3.4.1.2.1.

3.5 Marking.

3.5.1 Normal marking.- Marking shall be in accordance with Standard MIL-STD-129. The manufacturer's name, model number or type shall appear with the basic markings.

3.5.2 Precautionary handling markings.- Where instruments are internally blocked, braced, or supported for shipment, a warning label containing "WARNING DO NOT USE UNTIL INTERIOR BLOCKING IS REMOVED" shall be affixed to the equipment. Fragile labels and arrows shall be applied to the shipping container in accordance with Standard MIL-STD-129.

3.5.2.1 Shipping container shall be conspicuously marked in large letters:

DELICATE TEST EQUIPMENT
HANDLE WITH CARE

3.6 Workmanship.- All operations and processes involved in accomplishing the cleaning, packaging and packing requirements specified herein, shall be in accordance with the highest grade practices associated with this type of work. Caution shall be exercised in handling the item to preclude rough handling during packaging operation.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's quality assurance system.- The contractor shall provide and maintain an adequate quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.
(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

4.1.1 The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification and other applicable documents referenced herein.

4.2 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification shall consist of: (a) surveillance of the operations to determine whether the practices, methods and procedures of the written plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviations from the prescribed or agreed-upon procedures, or instances of poor practices which might have an affect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrections have been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

4.3 Contractor's inspection system.- Contractor's inspection and Government verification of contractor's inspection shall be in accordance with Specification MIL-I-45208.

4.4 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.4.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.5 Material. Control for requirements of material shall be in accordance with Specification MIL-I-45208.

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4.6 Tests.

4.6.1 Classification of defects.- The classification of defects in tables I and II shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any of all applicable requirements and to reject individual non-conforming items.

4.6.1.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be re-submitted for acceptance as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: None Defined

MAJOR: AQL 2.5%	Reqt Par.	Test Method
101. Proper blocking, bracing	3.2.1	Visual
102. Proper securing of clips and shields and presence of proper tubes	3.2.1	Visual
103. Proper protection of cables, probes, dividers and adapters	3.2.1	Visual
104. Component secured to instrument	3.2.1	Visual
105. Proper cleaning (Level A,C)	3.3.1.1, 3.3.2	MIL-P-116
106. Proper closure of fiberboard box, Group I, II, III (Level A,C)	3.3.1.2.1.1, 3.3.1.2.2.1, 3.3.1.2.3.1.5, 3.3.2	Visual
107. Proper padding of box within barrier, Group I, II (Level A,C)	3.3.1.2.1.2, 3.3.1.2.2.1, 3.3.2	Visual
108. Proper closure of wood containers, Group I, II, III, IV (Level A, C)	3.3.1.2.1.2, 3.3.1.2.1.2.2 3.3.1.2.3.3, 3.3.1.2.4.1 3.3.2	Visual
109. Proper immobilization Group III, IV (Level A, C)	3.3.1.2.3.1.2, 3.3.1.2.4.1, 3.3.2	Visual
110. Proper mounting Group III (Level A,C)	3.3.1.2.3.1.2, 3.3.2	Visual
111. Proper packing, Group I, II, III, IV (Level A, B, C)	3.4.1.1, 3.4.1.2, 3.4.1.2.1 3.4.2.1	Visual
112. Proper marking	3.5.1, 3.5.2, 3.5.2.1	Visual

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MINOR: AQL 6.5%	Reqt Par.	Test Method
201. Proper securing of cords	3.2.1	Visual
202. Proper clearance in pads, etc. for dial, knobs, and handles, Group I, II, III, IV (Level A,C)	3.3.1.2.1.1, 3.3.1.2.2.1 3.3.1.2.3, 3.3.1.2.4.1 3.3.2.	Visual
203. Proper thickness of fiberboard, Group I, II, III (Level A,C)	3.3.1.2.1.1, 3.3.1.2.2.1, 3.3.1.2.3.1.1, 3.3.2	Visual
204. Blunting of fiberboard boxes, Group I, II, III, (Level A,C)	3.3.1.2.1.1, 3.3.1.2.2.1, 3.3.1.2.3.2, 3.3.2	Visual
205. Proper clearance between unit and containers, Group III (Level A,C)	3.3.1.2.3.1.3, 3.3.1.2.3.3, 3.3.2	Visual
206. Proper wrapping, Group III IV (Level A,C)	3.3.1.2.3.1.4, 3.3.1.2.4.1 3.3.2	Visual
207. Proper quantity of unit packages, Group III, IV (Level A,B)	3.4.1.2	Visual
208. Weight limit, Group III, IV (Level A,B)	3.4.1.2	SME
209. Proper workmanship	3.6	Visual

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level I, Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%		
101. Quick leak, Group I, II, III (Level A)	3.3.1.2.1.1, 3.3.1.2.2.1 3.3.1.2.3.2, 3.3.1.2.4.1	MIL-P-116
102. Heat seal, Group I, II, III (Level A)	3.3.1.2.1.1, 3.3.1.2.2.1, 3.3.1.2.3.2, 3.3.1.2.4.1	MIL-P-116

MINOR: NONE DEFINED

4.7 The test methods outlined in MIL-P-116 shall be utilized to determine acceptability of those defects specifying MIL-P-116 for Test Method.

5. PREPARATION FOR DELIVERY
(Not applicable to this specification)

6. NOTES

6.1 Intended use.- Unless otherwise specified, this specification shall be used as packaging instructions for electronic test equipment, calibration standards and associated equipment. (See 6.5)

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable group number, figure number and level of packaging and packing.

6.3 Figures 1, 2, 3, 4, 5, 6, and 7 are examples of various packaging applications to be used for packaging electronic equipment. The selected application will depend on the configuration of the particular equipment to be packaged and will be determined by pilot pack.

6.4 Electronic test equipment to be returned by field sites to depots for calibration and repair should be retained until serviceable equipment is received for replacement. This procedure will facilitate the reuse of packaging material for return shipment.

6.5 The following list of items and the applicable group and figure numbers are for informational purposes only:

PART NO.	NOMENCLATURE	GROUPING	FIGURE
6616296	A. C. - D. C. Instrument Calbr Std RFL model 829	I	1
8616091	Frequency Counter, H.P. 524B	I	1
8616237	Oscilloscope Dual Input, H.P. model 150A	I	1
8616380	Power Supply, H.P. model 712B	I	1
8616087	Regulated Power Supply Pressman Assoc. model 3535R	I	1
Sig. Corp.	"S" Band Signal Generator HP616A	I	1
7659916	Signal Generator, Polarad Elec. Corp. (Microwave) Model MSG-1	I	1
7659917	Signal Generator, Polarad Elec. Corp. (Microwave) Model MSG-2	I	1
7659915	Signal Generator, Polarad Elec. Corp. (UHF) Model PMR	I	1
8616523	Stabilize Automatic Voltage Regulator Superior Electric Type EM-410R (110-120) (2 KVA)	I	1

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PART NO.	NOMENCLATURE	GROUPING	FIGURE
8616466	Amplifier & Null Detector GR Type 1231-BRFA	II	2
8616377	Audio Signal Generator, Hewlett-Packard Model 205AG	II	1
8616449	Bridge Oscillator GR Type 1330-A	II	1
8616191	Calibration Power Supply, John Fluke Model 406	II	1
8616511	Capacitance Bridge GR Type 716-CM	II	1
8616240	Electronic Counter, H.P. Model 523B	II	1
8616320	Freq Comparator, Lavoie Model 800	II	1
8616129	Heterodyne Freq. Meter P.R.D. Type 504	II	1
7659905	High Resistance Bridge, Mid-Eastern Elec. Inc. Model 801	II	1
8616446	Low-Distortion Oscillator GR Type 1301-A	II	2
8616395	Low Frequency Oscillator, H.P. Model 202C	II	1
7659904	Megatrometer, Mid-Eastern Electronics,Inc Mod. 700	II	1
8616139	Micromhometer, RCA Model WT-100A	II	1
8616462	Oscillator GR Model	II	2
8205525	Oscilloscope, Tektonix Type 515	II	1
8616188	Potentiometric D. C. Voltmeter, John Fluke Model 801	II	1
8616229	Power Meter, H.P., Model 430C	II	1
8616514	Power Supply, Beta Model 1005-2R-4	II	2
8614520	Power Supply, Beta Model 1010-2R	II	2
8616158	Scope & Spectrum Analyzer, Probescope Model SS-100 with Sweep Attachment	II	1
8616232 8616233	SHF Oscillator, H.P. Model 670H & Klystron Power Supply Model 717A	II	1
8616455	Signal Generator GR Model 1021-AU	II	1

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PART NO.	NOMENCLATURE	GROUPING	FIGURE
8616458	Signal Generator GR Model 1021-AW	II	1
7659914	Signal Generator, Measurements Corp. Model 80R	II	1
7659918	Signal Generator, Polarad Elec. Corp. Model MSG-34	II	1
3F3820.2	Signal Generator Sig Corps TS-588/U	II	1
8616211	SRI Corp Special UTS Universal Test Set, Code UNITEST	II	1
8616223	SRI Corp Model DYL Dynamometer Polyranger Code DYNFRE	II	1
8616469	Stabilize Automatic Voltage Regulator Superior Elec Type IE 51005	II	1
7659914	Test Oscillator, HP-650A	II	1
8205522	Time-Mark Generator, Tektronix Type 180-81	II	1
8616100	Transfer Oscillator, H.P. 540A	II	1
8616420	VHF Signal Generator, GR Type 1021-AV	II	1
Sig Corps	"X" Band Signal Generator HP 624C	II	1
7659895	AC DC Voltmeter, Electro Insts. Co. with Model DKX-020, Pre-Amp	III	4
8616417	Decade Capacitor GR Model 1419-A (.001 up - 1.11 uf)	III	4
8616452	Decade Resistor Gray Inst Type E-1044M	III	4
8616415	Decade Resistor GR Model 670-F	III	4
8616414	Decade Resistor Gray Inst Model E-1044 M4B	III	4
8616416	Decade Resistor Gray Inst Model E-1044-M4-3	III	4
8616451	Decade Resistor Gray Inst. E-1044 M4	III	4
8616453	Decade Resistor Gray Inst. Type E-1044M	III	4
8616350	Dynamometer Voltmeter, SRI Model D, Code DYVOSH	III	6

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PART NO.	NOMENCLATURE	GROUPING	FIGURE
8616362	Dynamometer Voltmeter, SRI Model D, Code DYVOCZ	III	6
8616133	Electronic Voltmeter, Ballantine Model 310A	III	3
8616094	Frequency Converters, H.P. 525A, 525B	III	3
8616243	Hetrodyne Frequency Meter, Gertach Model FM-3/PS-3	III	6
8616476	High Frequency V.T.V.M. H/P Model 410B	III	3
8616409	Kelvin-Wheatstone Bridge, Shallcross Model 638-R	III	6
8616226	Lab. Std, SRI Model B Code BAANKET	III	4
8616125	Microwave Freq. Meter Narda Model 802	III	6
7659902	Period Multiplier, Hewlett-Packard, Model HP526C	III	6
8616541	Portable Stand, Meter 7.5/15 Volt, Weston Model 341	III	6
8616544	Portable Stand, Meter 30/75 Volt Weston Model 341	III	6
8616547	Portable Stand Meter 75/150 Volt Weston Model 341	III	6
8616554	Portable Stand Meter 150/300 Volt Weston Model 341	III	6
8616557	Portable Stand Meter 300/600 Volt Weston Model 341	III	6
8616560	Portable Stand Meter 2.5/5 amps Weston Model 370	III	6
8616563	Portable Stand Meter 5/10 amps Weston Model 370		
8616566	Portable Stand Meter 75/150 MA Weston Model 370	III	6
8616569	Portable Stand Meter 150/300 MA Weston Model 370	III	6

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PART NO.	NOMENCLATURE	GROUPING	FIGURE
8616120	Power Supply for Unit Generator GR Type 1201A	III	3
8616472	Power Supply, Beta Model 9005-5	III	5
8616111	Precision Attenuator, H.P. Model J382A	III	5
8616112	Precision Attenuator, H.P. Model H382A	III	3
8205523	Precision Calibrator, Ballantine Model 420	III	3
8616258	Precision Frequency Meter, P.R.D. Model 583-D	III	3
8616270	Precision Frequency Meter, P.R.D. Model 578-A	III	3
8616273	Precision Frequency Meter, P.R.D. Model 579-A	III	3
8616331	Reference Std, SRI Model C, Code COLVY	III	6
8616334	Reference Std, SRI Model C, Code COLMUL	III	6
8616337	Reference Std, SRI Model C, Code CMVPO	III	6
8616340	Reference Std, SRI Model C, Code CILERA	III	6
8616345	Reference Std, SRI Model C, Code CICET	III	6
8616348	Reference Std, SRI Model C, Code CAMBI	III	6
8616194	SRI Corps Model B Lab Stand, Code BAVOMO	III	4
8616202	SRI Corps Model B Lab Stand, Code BAMILUK	III	4
8616310	SRI Corp Model C Portable Reference Stand, Code CICOL	III	6
8616317	SRI Corp Model C Portable Reference Stand, Code COLZO	III	6
8616326	SRI Corp Model DCHI d-c Kilo-voltmeter Code VOLTHI	III	4
8205515	Std Inductor, GR Type 1482-B	III	4

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PART NO.	NOMENCLATURE	GROUPING	FIGURE
8205518	Std Inductor, GR Type 1482-L	III	4
8205516	Std Inductor, GR Type 1462-E	III	4
8205517	Std Inductor, GR Type	III	4
8205519	Std Inductor, GR Type 1482-P	III	4
8205520	Std Inductor, GR Type 1482-T	III	4
7659896	Thermo-Ammeter, Greibach Inst. Co. Model 500	III	6
8616181	Time Freq. Calbr GR Model 1213-C	III	3
8205521	Time-Mark Generator, Tektronix Type 181-S1	III	3
7659901	Time Interval Unit, Hewlett-Packard, Model HP 526B	III	3
8616401	Universal Impedance Bridge, Electro Meas. Model 250-DA	III	6
8616383	Vacuum Tube Fork GR Model 723-D	III	4
8205524	Variac, GR Type V-10-MT	III	I
7659900	Video Amplifier, Hewlett-Packard, Model HP 526A	III	I
8616481	V.T.V.M. H/P Model 400H	III	3
8616389	Wheatstone Bridge, Gray Inst Model E-1023	III	4
8616350	Capacitor GR Type 505-A	IV	7
8616351	Capacitor GR Type 505-E	IV	7
8616352	Capacitor GR Type 505-F	IV	7
8616354	Capacitor GR Type 505-L	IV	7
8616355	Capacitor	IV	7
8616357	Capacitor GR Type 505-X	IV	7
7659903	Detector Mount, Hewlett-Packard, Model HP 440A	IV	7
8818102	Directional Coupler, F&R Type C610	IV	7

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PART NO.	NOMENCLATURE	GROUPING	FIGURE
8616103	Directional Coupler, R&R Type C610C	IV	7
8616104	Directional Coupler HP J762D	IV	7
8616105	Directional Coupler, F&R Type W610A	IV	7
8616106	Directional Coupler, F&R Type W610C	IV	7
8616107	Directional Coupler, F&R Type W610D	IV	7
8616493	Galvanometer 0.5 Div Weston Model 440	IV	7
8616261	Precision Frequency Meter, P.R.D. Model 585-A	IV	7
8616267	Precision Frequency Meter P.R.D. Model 587	IV	7
8616460	Range Extension Unit GR Model 1301-P	IV	7
7659897	Standard Capacitor, General Radio Co., Type 509L	IV	7
7659898	Standard Capacitor, General Radio Co., Type 509R	IV	7
7659899	Standard Capacitor, General Radio Co., Type 509X	IV	7
8205512	Std Capacitor, GR Type 509-K	IV	7
8205513	Std Capacitor, GR Type 509-M	IV	7
8205514	Std Capacitor, GR Type 509-T	IV	7
8616509	Std Capacitor, GR Type 509-Y	IV	7
8616373	Std Inductor (1 mh) GR Type 1481-A	IV	7
8616374	Std Inductor (5 mh) GR Type 1481-C	IV	7
8616429	Std Inductor (10 mh) GR Type 1481-D	IV	7
8616430	Std Inductor (50 mh), GR Type 1481-F	IV	7
8616431	Std Inductor (100 mh), GR Type 1481-G	IV	7
8616432	Std Inductor (1 oh), GR Type 1481-K	IV	7
7659906	Standard Resistors (Set), Mid-Eastern Electronics,Inc.	IV	7
8618289	Std Resistor (1:0 ohm) Gray Instr Model E-1243	IV	7

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PART NO.	NOMENCLATURE	GROUPING	FIGURE
8616290	Std Resistor (10.0 ohm), Gray Inst. Model E-1244	IV	7
8616291	Std Resistor (100.0 ohm), Gray Inst. E-1245	IV	7
8616292	Std Resistor (1K ohm) Gray Inst Model E-1246	IV	7
8616293	Std Resistor (10K ohm) Gray Inst Model E-1247	IV	7
8616294	Std Resistor (0.1 ohm), Gray Inst. (Reinchsanstalf Type) Model E-1230	IV	7
8616328	Std Resistor (50K ohm), GR Type 500-T	IV	7
8616329	Std Resistor (100K ohms) GR Type 500-U	IV	7
8616406	Std Resistor (500K Ohms) GR Type 500-W	IV	7
8616407	Std Resistor (1 meg) GR Type 500-X	IV	7
8603811	Thermistor Mount, H.P. Model 477B (10 MC - 10 KMC)	IV	7
8616108	W. G. Thermistor Mount, H.P. Model G487A	IV	7
8616109	W. C. Thermistor Mount, H.P. Model J487A	IV	7
8616110	W. C. Thermistor Mount, H.P. Model H487A	IV	7

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1. Electronic Test Unit
2. Fiberboard sleeves, panels, fillers, die cuts and pads.*
   - 2a. Bottom Panel (with filler pads as req'd.)
   - 2b. Side pads (die cuts) - as req'd.
   - 2c. Front and back panels (with filler pads as req'd.)
   - 2d. Sleeve
   - 2e. Top panel

*Filler pads, cells, etc. may be secured to fiberboard panels with pressure sensitive tape conforming to Spec. PPP-T-76 or adhesive conforming to Spec. MIL-A-101.

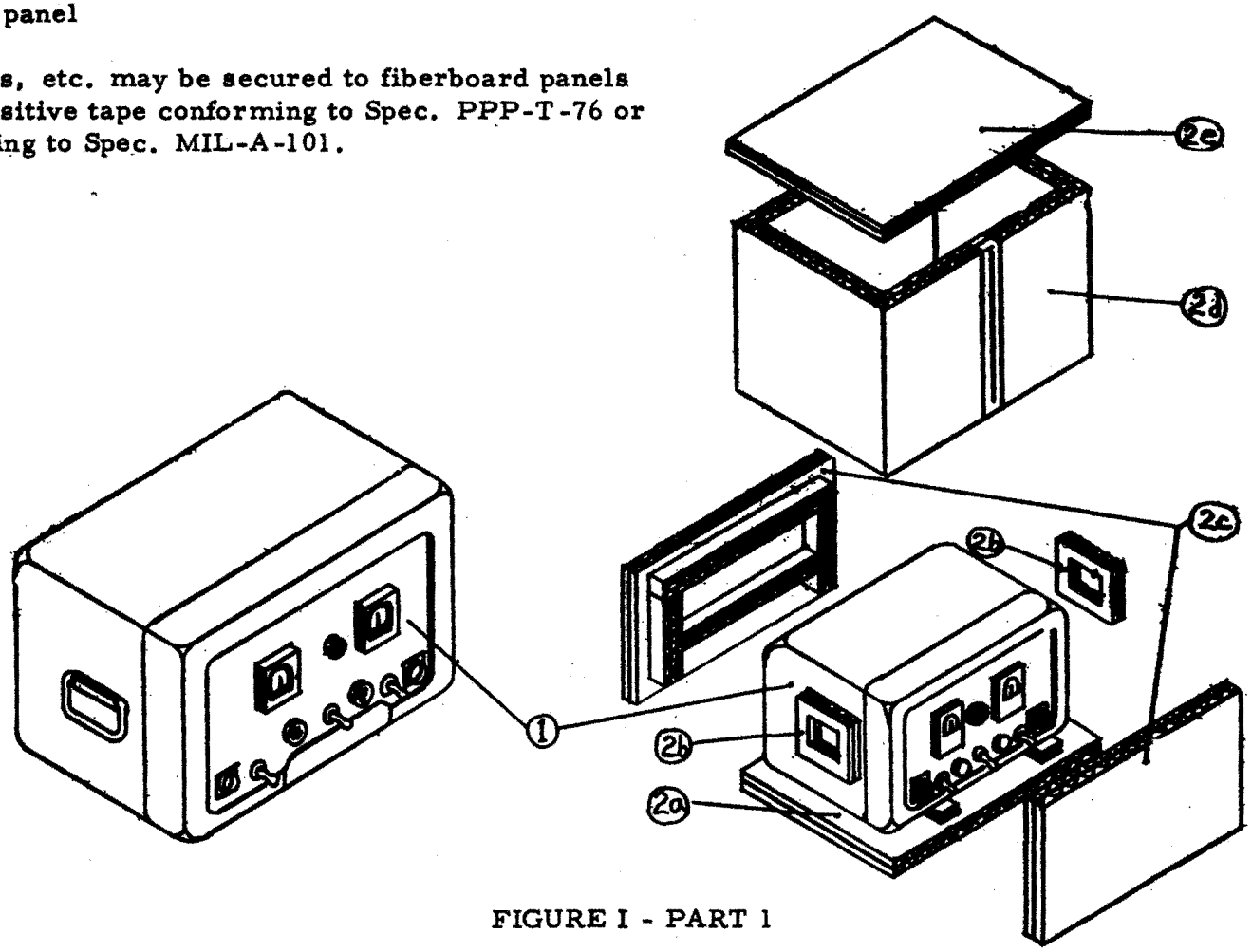


FIGURE I - PART 1

PACKAGING DETAIL FOR ELECTRONIC TEST EQUIPMENT (CABINET MOUNTED)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:44Z
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1. Electronic Test Unit.
2. Fiberboard panels, reinforced and taped.
   - 2a. Reinforcement strips - pressure sensitive tape, type III of PPP-T-97
   - 2b. Tape - pressure-sensitive - PPP-T-76 - 2" wide
3. Vaporproof barrier material - class I, MIL-B-131.
4. Bound cushioning material, type IV - MIL-C-7769.
5. Shipping container.

NOTE: Closure of wood shipping container shall be with slotted wood screws to facilitate reuse.

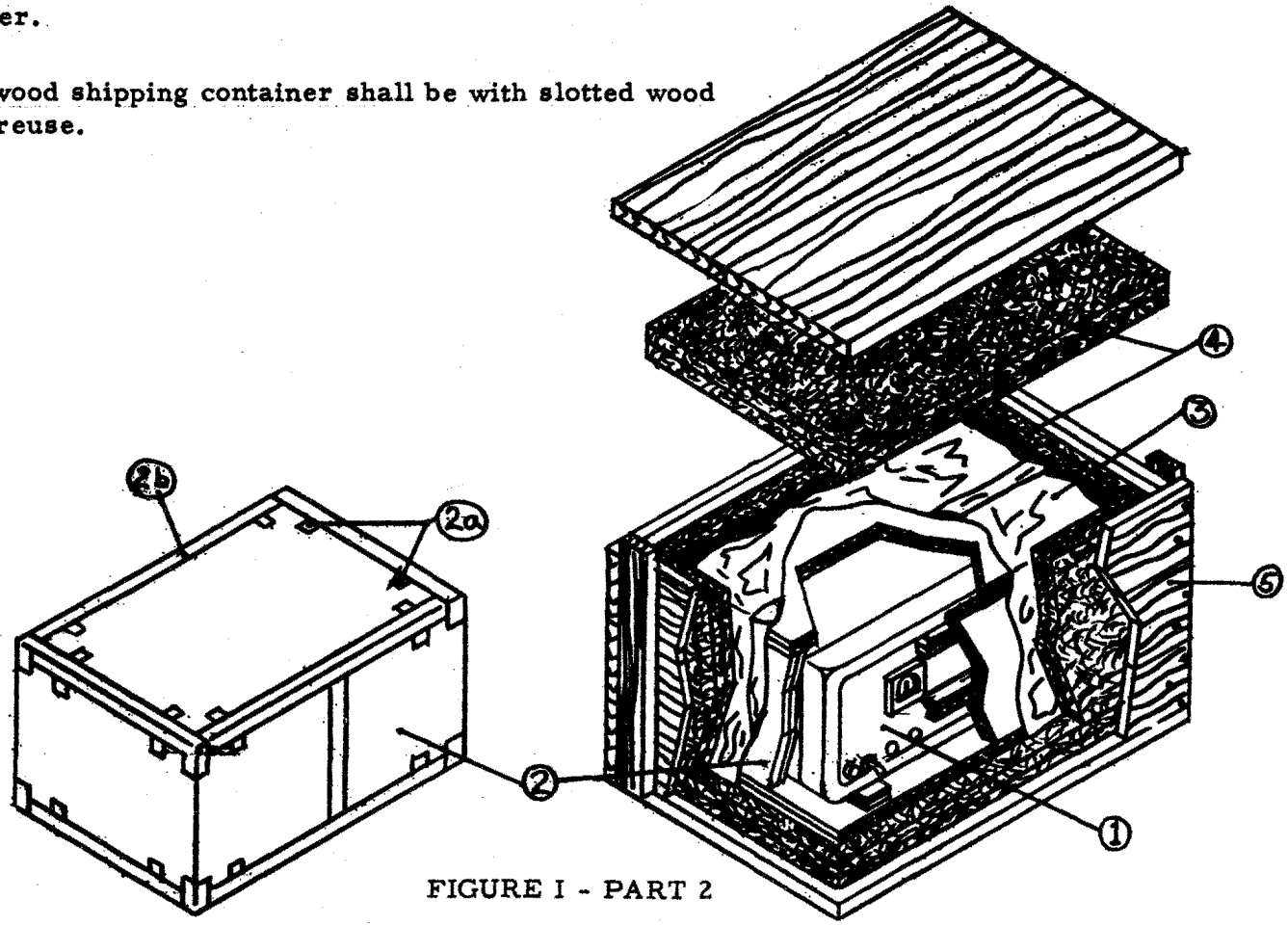


FIGURE I - PART 2

PACKAGING DETAIL FOR ELECTRONIC TEST EQUIPMENT (CABINET MOUNTED)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:44Z
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1. Electronic Test Unit.
2. Fiberboard panels and filler pads.*
   - 2a. Bottom Panel - with filler pads (as required)
   - 2b. 3/4 Sleeve with filler pads (as required)
   - 2c. Top panel with filler pads (as required)
3. Cleated plywood front panel with felt pads, plywood conforming to type II or III-NN-P-515.
4. Hardware - as required.

*Filler pads, die cuts, cells, etc. may be secured to fiberboard panels with pressure sensitive tape conforming to Spec. PPP-T-76 or adhesive conforming to Spec. MIL-A-101.

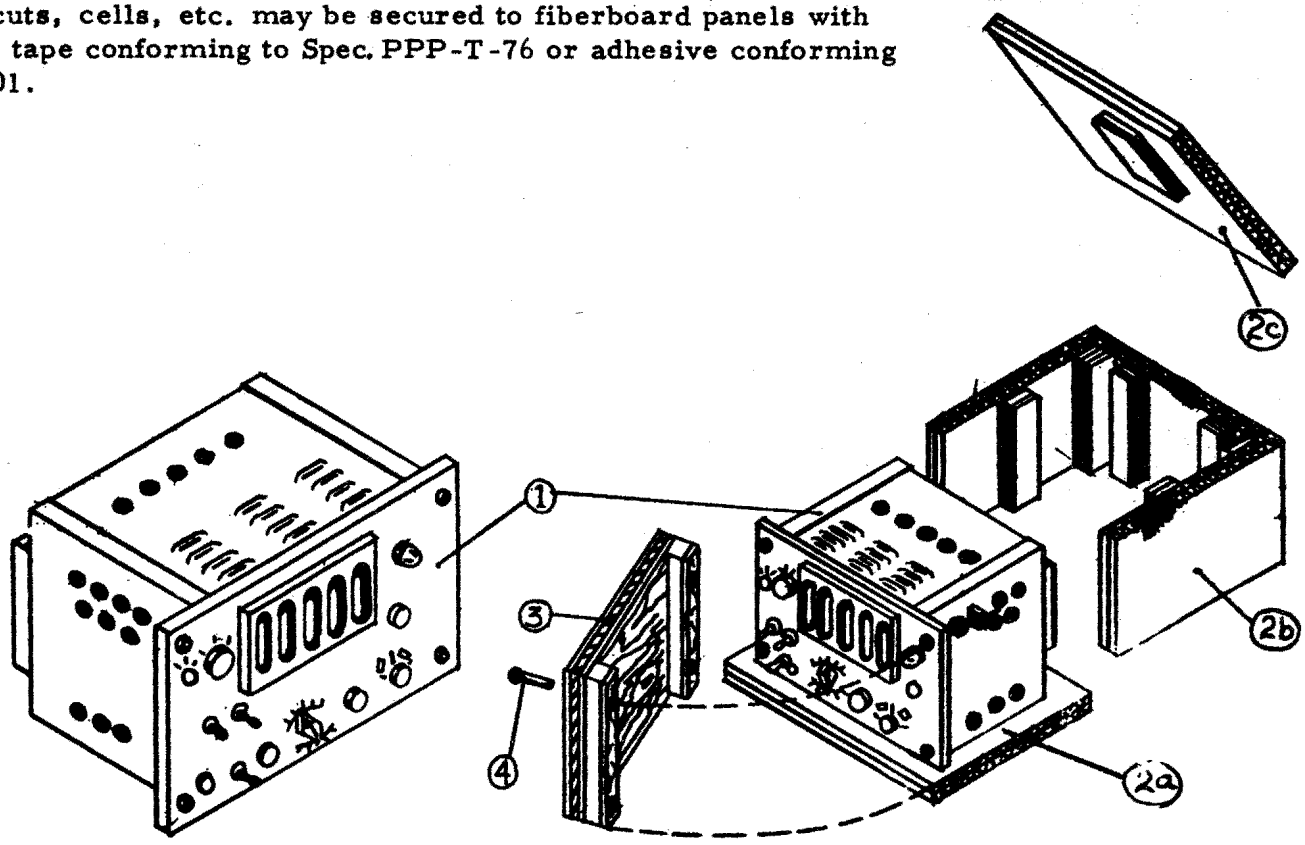


FIGURE 2 - PART 1

PACKAGING DETAIL FOR ELECTRONIC TEST EQUIPMENT (RACK MOUNTED)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:44Z
Check the source to verify that this is the current version before use.

1. Electronic Test Unit
2. Panels, reinforced and taped.
    2a. Reinforcement strips - pressure sensitive tape, type III, PPP-T-97
    2b. Tape - pressure sensitive - PPP-T-76 (2" wide)
3. Vaporproof barrier material, class 1, MIL-B-131..
4. Bound cushioning material, type IV, MIL-C-7769.
5. Shipping container.

NOTE: Closure of wood shipping container shall be with slotted wood screws, to facilitate reuse.

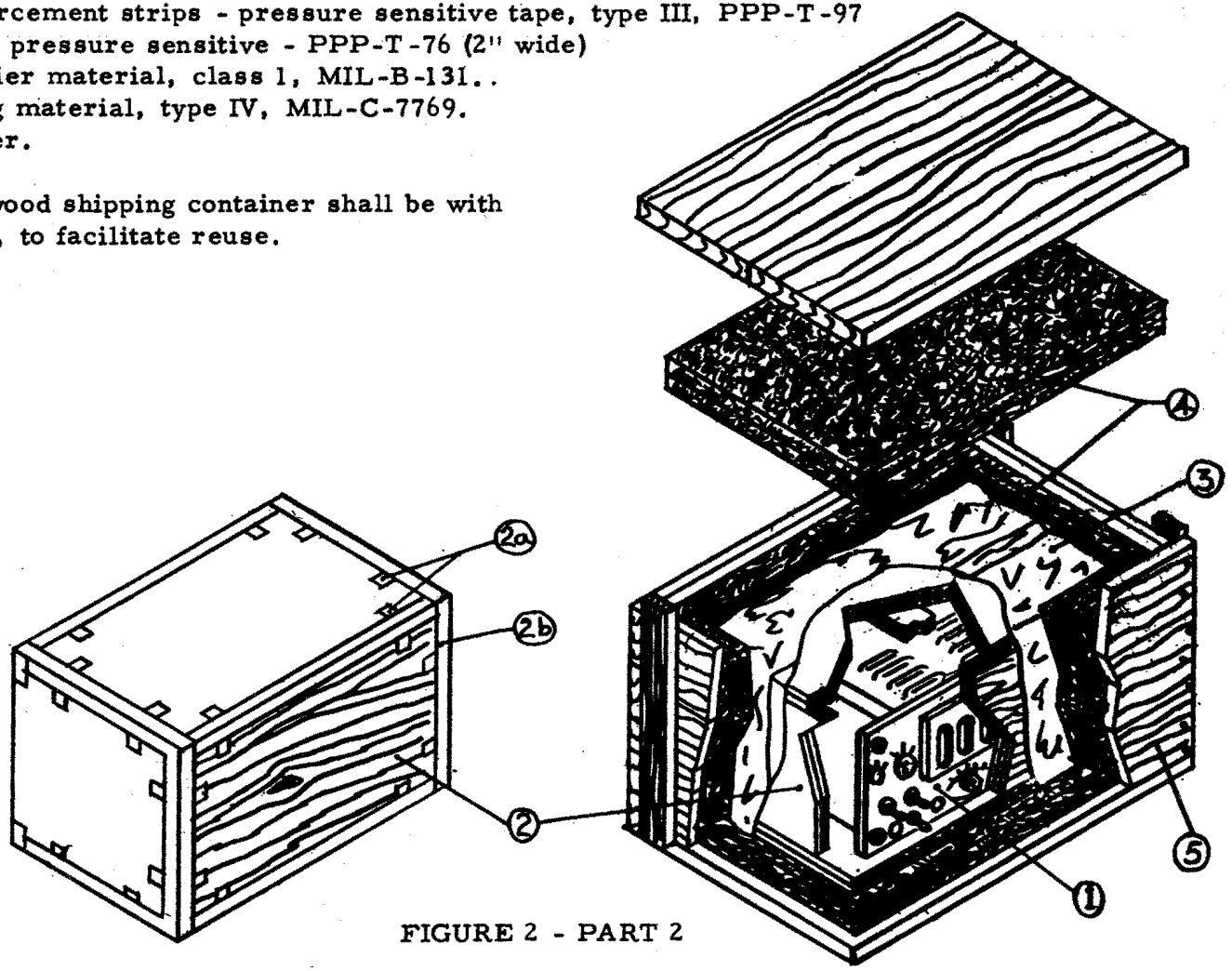


FIGURE 2 - PART 2

PACKAGING DETAIL FOR ELECTRONIC TEST EQUIPMENT (RACK MOUNTED)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:44Z
Check the source to verify that this is the current version before use.

1. Electronic Test Equipment.
2. Fiberboard sleeve, panels, fillers, die cuts and pads.
   - 2a. Bottom panel (with filler pads as required)
   - 2b. Front panel (with filler pads as required)
   - 2c. Sleeve (with filler pads as required)
   - 2d. Top pad (with filler pads as required)

NOTE: Filler pads, cells, etc shall be secured to fiberboard panels with pressure sensitive tape conforming to Spec. PPP-T-76 or adhesive conforming to Spec. MIL-A-101.

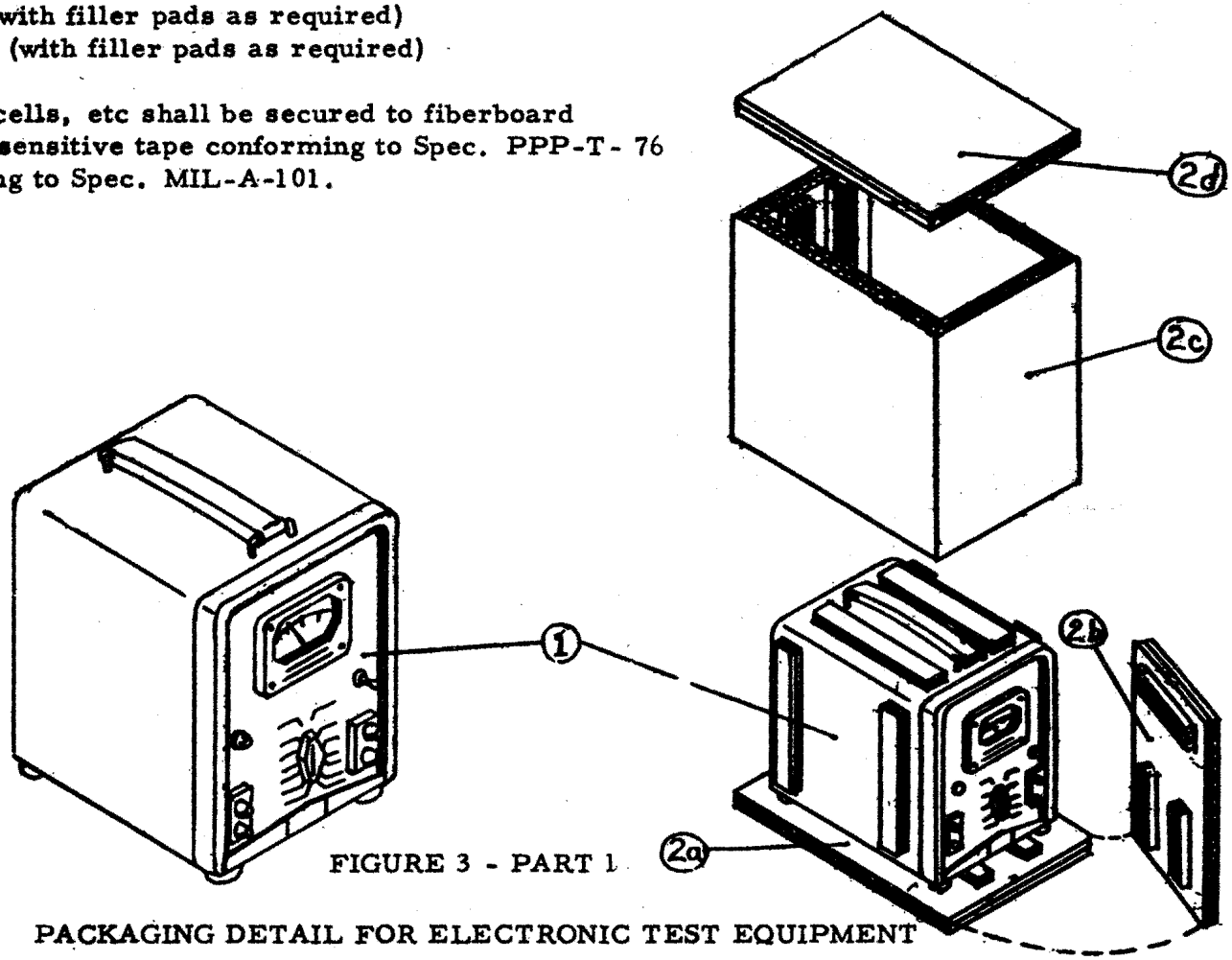


FIGURE 3 - PART 1

PACKAGING DETAIL FOR ELECTRONIC TEST EQUIPMENT

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:44Z
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1. Electronic Test Equipment.
2. Fiberboard panels reinforced and taped.
   2a. Reinforcement strips - pressure sensitive tape, type III, PPP-T-97.
   2b. Tape pressure sensitive - Spec. PPP-T- 76 (2 inches wide)
3. Vaporproof barrier material, class I, MIL-B-131.
4. Bound cushioning material, type IV, MIL-C-7769 (min. 2" thickness)
5. Fiberboard container.

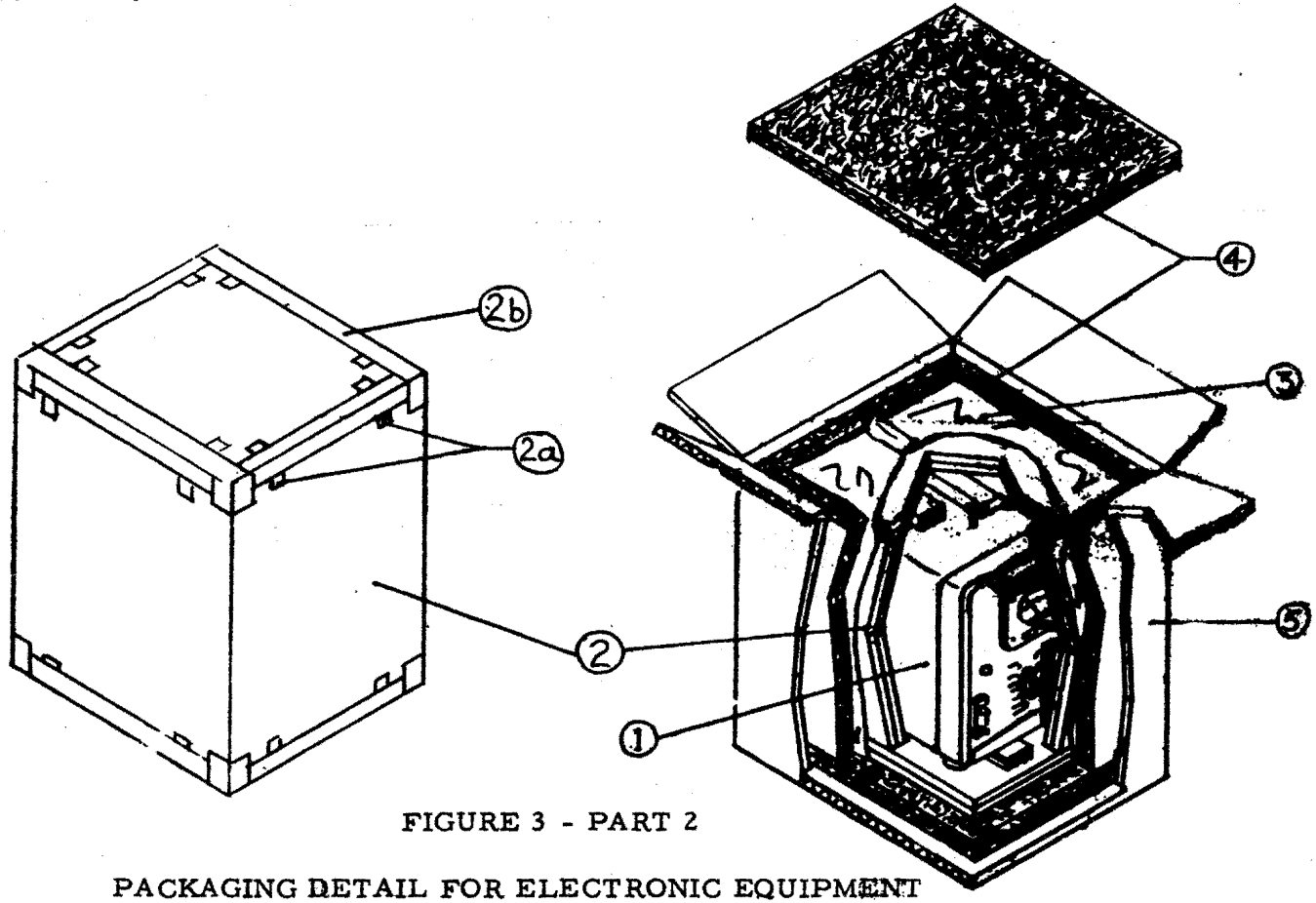


FIGURE 3 - PART 2

PACKAGING DETAIL FOR ELECTRONIC EQUIPMENT

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:44Z
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1. Electronic Test Equipment.
2. Inner fiberboard container, type I, class 1, style RSC, Spec. PPP-B-636.
3. Hold down liner, double wall, type I, class 1, Spec. PPP-B-636.
4. Vaporproof barrier material, class I, Spec. MIL-B-121.
5. Bound cushioning material, type IV of Spec. MIL-C-7769 - (Min. 2 inch thickness)
6. Outer container, type I, class 2, style RSC of Spec. PPP-B-636.

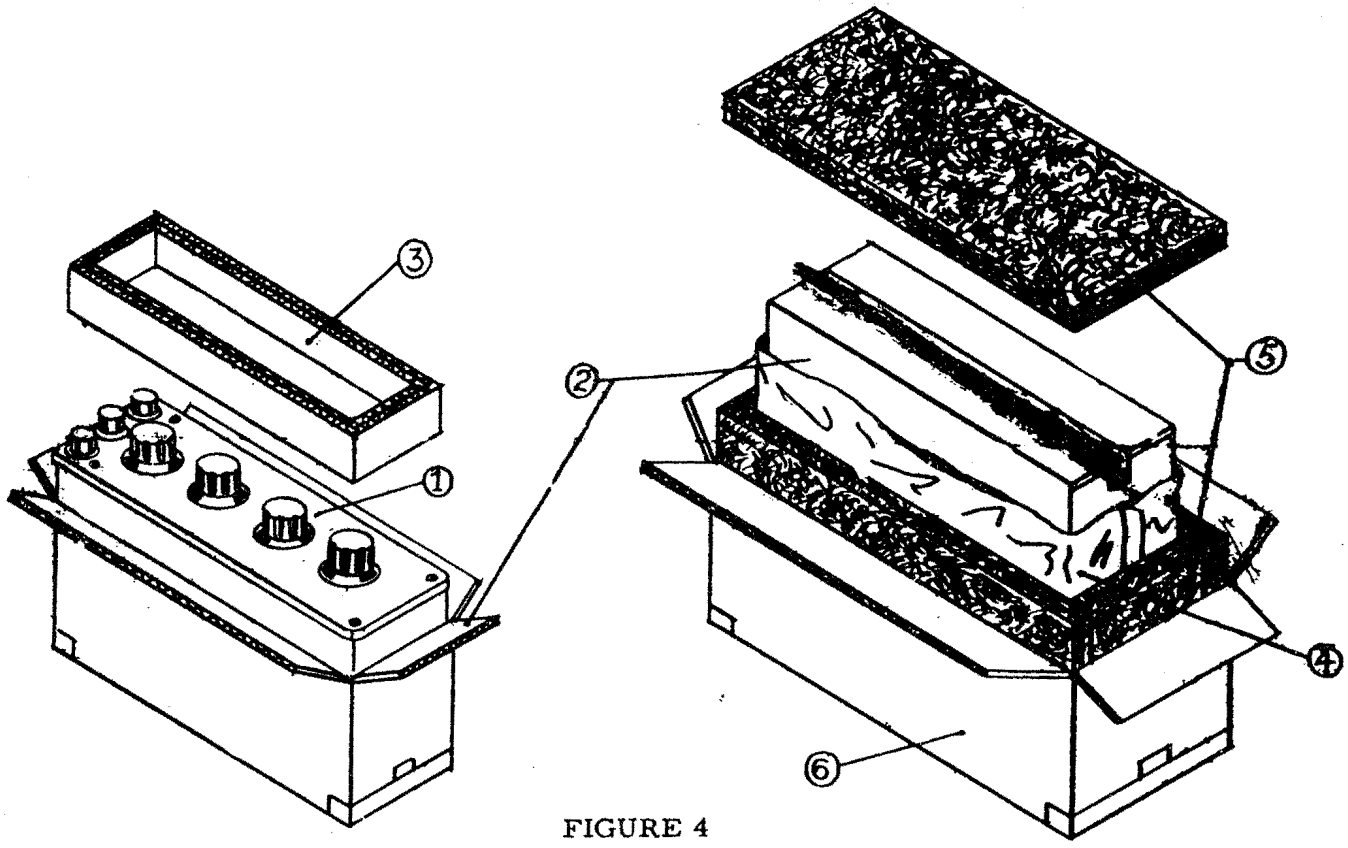


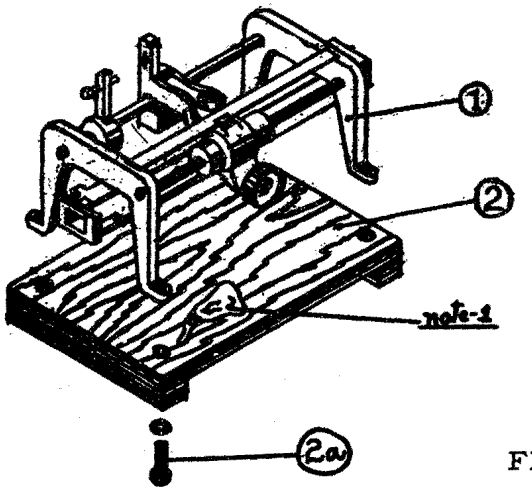
FIGURE 4

PACKAGING DETAIL FOR ELECTRONIC TEST EQUIPMENT

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:44Z
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1. Electronic Test Equipment.
2. Cleated plywood base - plywood type II or III of NN-P-515.
   2a. Hardware, as req'd. to secure unit to base.
3. Inner fiberboard box, type I, class 1, style RSC of Spec. PPP-B-636.
4. Hold down liner - double wall, type I, class 1 of Spec. PPP-B-636.
5. Vaporproof barrier material, class I of Spec. MIL-B-131.
6. Bound cushioning material - type IV of Spec. MIL-C-7769 (min. 2" thick)
7. Outer fiberboard box, type I, class 2, style RSC of Spec. PPP-B-636.

NOTE: Leveling screws shall be removed from unit and placed in a draw string cotton cloth bag. Secure bag to unit.



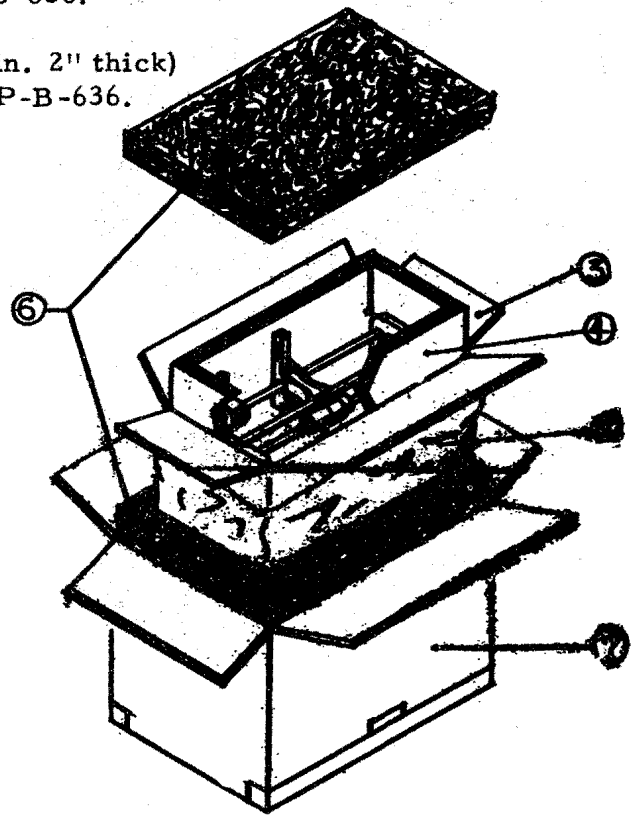


FIGURE 5

PACKAGING DETAIL FOR ELECTRONIC TEST EQUIPMENT

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26.

1. Electronic Test Equipment.
2. Wrap - single face corrugated fiberboard - PPP-P-291.
3. Vaporproof barrier material, class I, MIL-B-131.
4. Bound cushioning material, type IV of MIL-C-7769 (min. 2" thickness)
5. Outer fiberboard box, type I, class 2, style RSC of Spec. PPP-B-636.

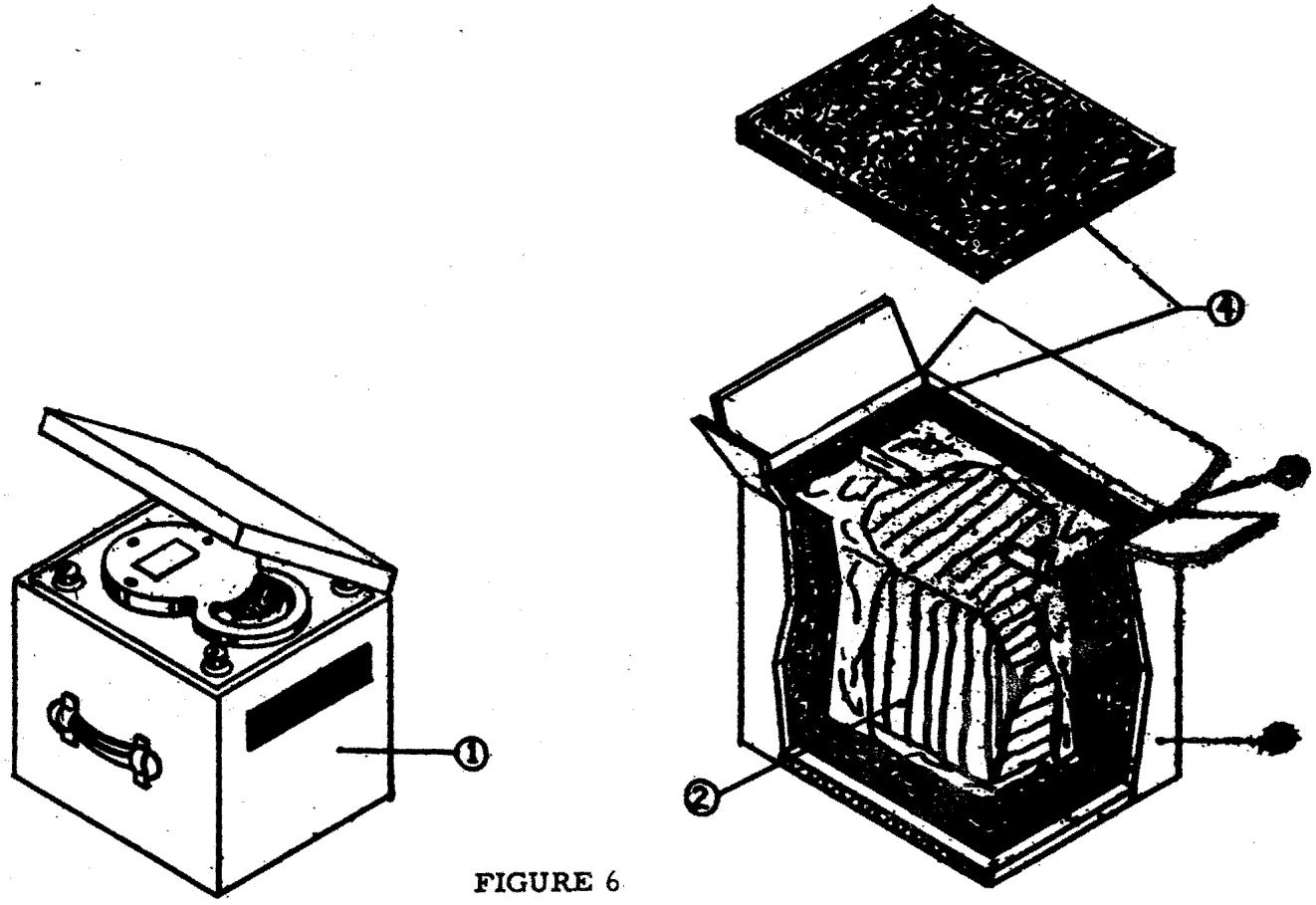


FIGURE 6

PACKAGING DETAIL FOR ELECTRONIC TEST EQUIPMENT

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:44Z
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1. Electronic Test Equipment.
2. Barrier material, class 1, grade A, MIL-B-121.
3. Cushioning material, cellulosic, type III, class B, PPP-C-843 (min. 1/2 inch thick)
4. Vaporproof barrier material, class 1, MIL-B-131.
5. Fiberboard box, type 1, class 2, W5c, style RSC or OPF of Spec. PPP-B-636.

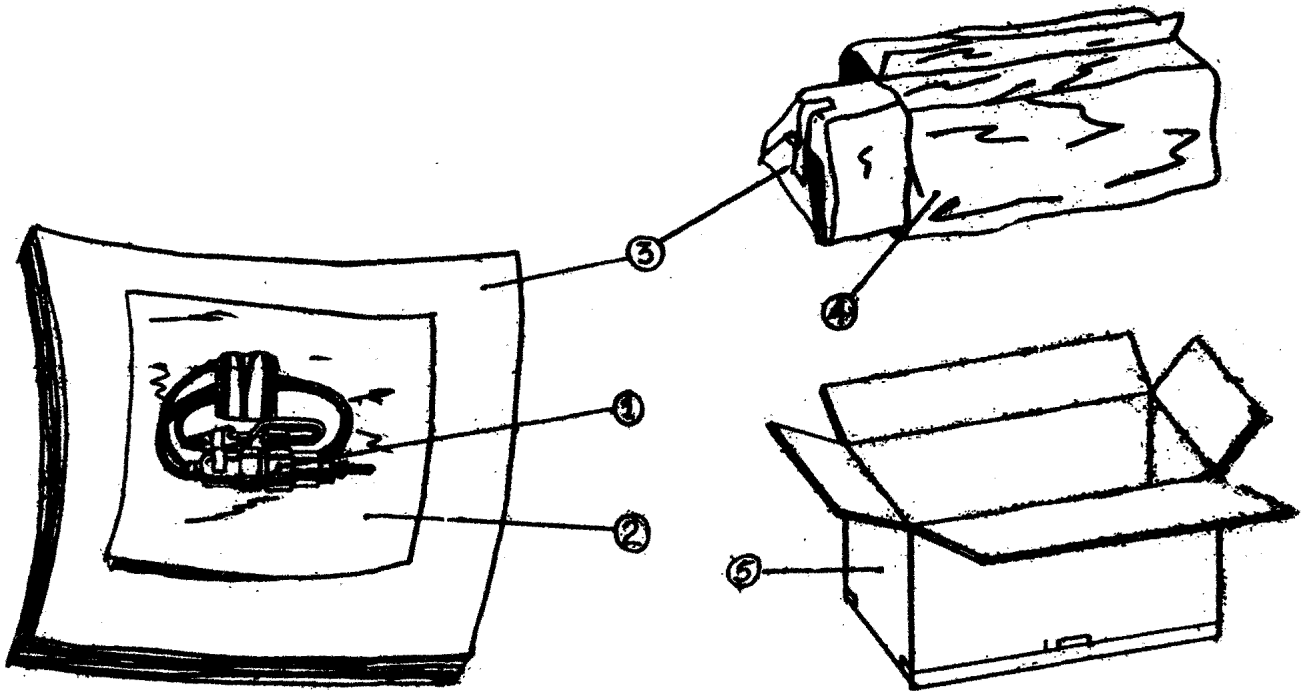


FIGURE 7

PACKAGING DETAIL FOR ELECTRONIC TEST EQUIPMENT

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